

Aarya Mehta

aarya.m.mehta@gmail.com | 972-834-6821 | Frisco, TX | US Citizen

Education:

Purdue University
Aerospace Engineering
GPA: 3.93/4.00

May 2029
West Lafayette, IN

Skills: Python, SQL, MATLAB, C, Java, Data Analysis, Microsoft Excel, SolidWorks, Fusion 360, NX 2306, Ansys Workbench, Microsoft Office Suite

Experience:

Mechanical R&D Engineering Intern

May 2026 - August 2026

SharkNinja, Frozen Treats Team

- Advanced a next-generation frozen treat product from experimental development to a production-representative prototype through mechanical design, quantitative testing, data analysis, and iterative optimization.
- Executed hundreds of structured performance tests across 3 prototype iterations, developing testing methodologies around critical performance metrics and analyzing results to identify optimization opportunities; contributed to a ~70% reduction in processing time while improving product performance and consistency.
- Developed a data-driven framework to evaluate product performance across 100+ tests, validating ~15 existing KPIs and analyzing voltage, current, torque, power, temperature, processing time, and product characteristics to inform design decisions.
- Built an Arduino-based data acquisition dashboard to automate collection and organization of test data, enabling more efficient analysis across repeated experiments.
- Led NTC thermistor calibration by developing a test rig and experimental test plan, collecting and analyzing hundreds of data points, and creating a resistance-temperature correlation that reduced measurement error from ~4°C to ~0.1°C.
- Designed 6 CAD concepts and iterated through 30+ 3D-printed prototypes, using reverse engineering, experimental results, and comparative evaluation to select and refine design solutions.

LOx Inducer Manufacturing Responsible Engineer

September 2025 - Present

Purdue Undergraduate Rocket Propulsion Lab (PURPL)

- Lead the LOx inducer hardware workstream for Marlin, a 5000 lbf turbopump rocket engine, applying quantitative analysis and engineering trade studies to guide design and manufacturing decisions.
- Completed a materials trade study evaluating identified performance and design parameters to recommend 316L stainless steel for the LOx inducer.
- Verified design viability through first-principles calculations and FEA, evaluating structural safety margins and thermal expansion before manufacturing.
- Created CAM toolpaths and machined complex inducer geometry on a 5-axis UMC 500 CNC mill, translating analytical design decisions into physical hardware.

Researcher/Senior Mentor

October 2023 - March 2025

NASA High School Aerospace Scholars (HAS)

- Collaborated with NASA consultants and peers to develop and present an Artemis Gateway expansion mission proposal, integrating CAD modeling, coding, engineering design, risk mitigation, and exploration planning.
- Selected as Senior Mentor, providing technical and leadership guidance to junior scholars while completing training in communication and project management.

Projects:

Dark Matter Research | Python

July 2023 - May 2025

- Co-authored research estimating dark matter fractions in early-type galaxies using the Broken-Power-Law model and machine learning techniques, applying Python-based analysis to astronomical datasets.
- Contributed to research published in Monthly Notices of the Royal Astronomical Society (DOI: 10.1093/mnras/staf811), demonstrating experience communicating quantitative findings through peer-reviewed research.

Interests: Tennis, Basketball, Climbing, Astrophotography